

Research on Additional Image Sources for Cambodian 20,000 and 50,000 Riel Banknotes

Scope and key findings

The National Bank of Cambodia page for banknotes in circulation currently shows **three 20,000-riel versions** and **three 50,000-riel versions** still in circulation. For **20,000 riels**, the page lists issues from **1995**, **2008**, and **2018**. Collector references reconcile that newest note as a **2017-dated note issued into circulation in May 2018**. For **50,000 riels**, the page lists issues from **1995**, **2001**, and **2014**, while collector references identify the newest design as a **2013-dated note issued in May 2014**. ¹

For your specific use case, the strongest immediately useful public sources are **Numista** for clean front/back note art and **Roboflow** for real object-detection scene images. The weakest sources for direct training are the **official National Bank of Cambodia images**, because the image files I checked are specimen-style images with **SPECIMEN overlays and zero serials**, and **Wikimedia Commons**, where Cambodian banknote files are currently being nominated for deletion on copyright grounds. ²

The practical bottleneck is not basic note art anymore. It is **licensable, non-specimen, front/back coverage for old designs**, plus **real phone-photo variation** for the weak classes. The biggest gaps I found are **KHR_20000 1995** and, to a lesser extent, **KHR_50000 2001**. ³

Version inventory

For model-building purposes, I would treat the banknotes as these visual design families:

Denomination	Design family I found	Status
20,000 riels	1995 old design	Still listed by NBC as circulating. ⁴
20,000 riels	2008 design	Still listed by NBC as circulating. ⁴
20,000 riels	2017-dated note issued in 2018	Current newest design; Numista catalogs it as a circulating commemorative tied to the King's 65th birthday. ⁵
50,000 riels	1995-style design family	NBC lists a 1995 issue; Numista records both ND 1995 and 1998 variants within the same P-49 design family. ⁶
50,000 riels	2001 design	Still listed by NBC as circulating. ⁷
50,000 riels	2013-dated note issued in 2014	Current newest design. ⁸

The main modeling implication is that the catalog year is not always the circulation year. For the newest notes, **"2017" vs. "2018"** on 20,000 riels and **"2013" vs. "2014"** on 50,000 riels are not contradictory sources; they are two ways of referring to the same design: the note's printed/catalog year versus the circulation issue date. ¹

Source table for KHR_20000

Quality ratings below are about usefulness for **note-level object detection**, not collector-grade microprinting study.

Version or coverage	Source	URL	Front/back pair	Quality and exposed resolution	Visible issues
All three 20,000-riel versions on one authoritative page: 1995, 2008, 2017-dated issued 2018	National Bank of Cambodia banknotes gallery	https://www.nbc.gov.kh/english/about_the_bank/banknotes_in_circulation.php	Yes, per version	Good official color; medium web JPGs	Official image check shows specific style artifacts such as SPEC text and zero no real clutter phone photo condition
2008 standard circulation note	Numista	https://en.numista.com/206056	Yes	Good ; clean real-note front/back scans; web image size not stated publicly on page	White border back scan or scan like a flat scan rather than phone

Version or coverage	Source	URL	Front/back pair	Quality and exposed resolution	Visible issues
2017-dated note issued in 2018	Numista	https://en.numista.com/208877	Yes	Good ; clean front/back scans; web image size not stated publicly	Slight creases; scan still fl domain scene domain
1995 old design	Numista	https://en.numista.com/217619	Yes	Fair ; older scans	Visible cgb.f wate dust, scan artifa weak clean for th versi
2008 standard circulation note	Banknote World Educational	https://www.banknoteworld.org/cambodia-20-000-riels-2008-p-60a.html	Yes	Good catalog scan; page states scan donated by Colnect	Flat s only; scene varie publi state open
2017-dated newest 20,000-riel note	Banknote World Educational	https://www.banknoteworld.org/cambodia-20-000-riels-2017-p-70.html	Yes	Good catalog scan; page states scan donated by Colnect	Flat s only; scene varia

Version or coverage	Source	URL	Front/back pair	Quality and exposed resolution	Visible issues
2008 note in marketplace photos	eBay listing	https://www.ebay.com/itm/236235522941	Likely yes; the listing exposes 3 pictures	Good seller-photo quality	Product photo domain fully wild; rights not o
Mixed denomination scene dataset including 20000_Riel	Roboflow Universe	https://universe.roboflow.com/khmer-riel-classification-computer-vision/cambodia-currency-project	No; scene images, not note pairs	Best direct-training source among public options; class exists; dataset version shows 615 total images , split 537/51/27 , with preprocessing to 640×640 and standard augmentations	Public meta does expo versio coun you r manu inspe whet your class conta 2008 the 2 desig both
Mixed Cambodian banknote scene dataset	Kaggle	https://www.kaggle.com/datasets/thychantha/khmer-currency-itc-ams-gen2-c	No; scene images, not note pairs	Public description says 10 distinct classes with different lighting, angles, and backgrounds	Public meta does expo 20,00 speci coun versio manu review need

Version or coverage	Source	URL	Front/back pair	Quality and exposed resolution	Visible issues
2017 20,000-riel file on Wikimedia Commons	Wikimedia Commons	https://commons.wikimedia.org/wiki/File:Cambodia_20000_riel_2017_avers.jpg	Yes, with matching reverse file	Public preview mentions 800 × 374 with 1,280 × 598 original	Common file is deleted nom and r Camb bank files a flagg copy conce

The usable 20,000-riel picture is therefore fairly clear: **Numista 2008** is your best open clean template, **Numista 2017/P-70** is useful if your project really is academic/noncommercial, and **Roboflow** is the fastest public path to true scene images. The hard hole is still **1995**. ¹⁹

Source table for KHR_50000

Version or coverage	Source	URL	Front/back pair	Quality and exposed resolution	Visible issues
All three 50,000-riel versions on one authoritative page: 1995, 2001, 2013-dated issued 2014	National Bank of Cambodia banknotes gallery	https://www.nbc.gov.kh/english/about_the_bank/banknotes_in_circulation.php	Yes, per version	Good official color; medium web JPGs	Official images checked are specimen-style with SPECIMEN/zero-serial issues and no real wear or scene variation
2013-dated note issued in 2014	Numista	https://en.numista.com/215642	Yes	Good ; clean scans; web image size not stated publicly	Slight center fold and scan-domain flatness, but otherwise very usable

Version or coverage	Source	URL	Front/ back pair	Quality and exposed resolution	Visible issues
2001 design	Numista	https://en.numista.com/207992	Yes	Good ; visually clean front/ back pair	Flat scan only; page does not surface an explicit CC license in the snippet I could verify
1995-style design family, represented by a 1998 variant in Numista	Numista	https://en.numista.com/211651	Yes	Fair ; worn but usable	Aging, stains, fold lines, and lower contrast; still a valuable old-design source
2013-dated / 2014-issued note	Banknotes.com	https://www.banknotes.com/cambodia-50000-riels-2013.html	Yes, front/ back sample image	Medium-good web catalog image; dimensions and note metadata are exposed on page	Preview image is commercial/sample-style and visibly watermarked in the search preview
1995 ND old-style note	Banknote World Educational	https://www.banknoteworld.org/cambodia-50-000-riels-1995-nd-p-49.html	Yes, front/ back combined page image	Medium	Commercial scan; flat-scan domain; not scene data
2013 note in marketplace photos	eBay listing	https://www.ebay.com/itm/387341312785	No; page exposes 1 image only	Usable real note photo , but limited	Only one view; limited training diversity; not openly licensed

Version or coverage	Source	URL	Front/back pair	Quality and exposed resolution	Visible issues
Real angled photo of old 50,000-riel reverse	Alamy stock photo	https://www.alamy.com/a-fifty-thousand-riels-banknote-from-cambodia-showing-the-road-leading-to-preah-vihear-temple-used-banknote-photographed-at-an-angle-image401973871.html	No; reverse only in the surfaced result	Useful because it is an actual used note photographed at an angle	Single-side only; not free; stock-photo licensing terms must be checked for ML use
Mixed denomination scene dataset including 50000_Riel	Roboflow Universe	https://universe.roboflow.com/khmer-riel-classification-computer-vision/cambodia-currency-project	No; scene images	Best direct-training source among public options ; UI exposes 50000_Riel class and downloadable dataset formats	Same caveat as 20,000: public metadata does not reveal per-version composition for the 50,000 class
Mixed Cambodian banknote scene dataset	Kaggle	https://www.kaggle.com/datasets/thychantha/khmer-currency-itc-ams-gen2-c	No; scene images	Public description says 10 classes under varied lighting, angles, and backgrounds	Per-denomination counts and version split are not exposed publicly

For 50,000 riels, the picture is stronger than for 20,000 riels. You already have one very good open clean source for the newest design (**Numista 2013, CC BY-SA**) and one open but worn source for the oldest design (**Numista 1998 variant of the 1995-style series, CC0**). The weakest licensing point is the **2001 design**, where I found a clean pair but not a clearly open reuse grant. ²⁸

Missing versions and download priority

The clearest remaining gaps are these:

- **KHR_20000 1995** is still the hardest version. I found front/back images, but the strongest easily accessible pair is on Numista with **CGB attribution/watermark**, not on an openly licensed clean page. ¹²
- **KHR_20000 2017/P-70** is visually well covered, but the best clean public pair I found is **CC BY-NC**, which is fine for an academic prototype and awkward for a future commercial/open dataset. ²⁹
- **KHR_50000 2001** has a clean Numista pair, but I could not verify an explicit open license on the page the way I could for the 2013 and 1998/1995-style entries. ³⁰

- Truly **in-the-wild phone-photo examples** are still thin. The best public scene source is Roboflow; Kaggle may help, but both need manual checks for version composition inside the 20,000 and 50,000 classes. ³¹

I also checked two catalog sources that are worth manual follow-up even though they were not reliable download targets in this session. **BanknoteDB** clearly has Cambodia issuer pages and at least a **50,000 Riels 2013** entry in search results, but it was behind verification/anti-bot pages when I attempted to use it. The **Beyond Angkor / Currency of Cambodia** collector wiki surfaced relevant category pages for **20,000** and **50,000** riels, but direct fetches returned errors or 403 responses here. ³²

Some travel/blog pages exist, but they are poor priorities. The Blogspot post about the 2008 20,000-riel note is just a low-resolution repost of the official image, and the traveler currency guide shows small illustrative images rather than dataset-grade photography. ³³

The most practical first downloads are:

1. **Roboflow Cambodia Currency Project** first, because it is already an object-detection dataset, it explicitly contains `20000_Riel` and `50000_Riel`, it is downloadable, and it is under **CC BY 4.0**. Start by auditing how many 20,000 and 50,000 examples it actually contains and which design families appear in those classes. ¹⁶
2. **Numista 20,000 riels 2008, Numista 50,000 riels 2013, and Numista 50,000 riels 1998/1995-style** next. Those are your best public clean-template set, and two of the three have clearly favorable licenses (**CC BY-SA** and **CC0**). ³⁴
3. **Numista 20,000 riels 2017/P-70** after that, if your use really remains academic/noncommercial. It is probably your fastest way to stabilize the newest 20,000-riel class without waiting for your own photos. ²⁹
4. Use **eBay, Alamy, Banknote World, and Banknotes.com** only as gap-fillers, permission targets, or reference sources. They are useful for realism and old-design coverage, but they are not the clean legal backbone of a redistributable dataset. ³⁵

Conclusion on whether real data is still needed

Synthetic augmentation will help a lot here, because banknotes are flat, highly structured objects and you now have several usable front/back sources for template extraction. That is especially true for **20,000 riels 2008, 50,000 riels 2013, and 50,000 riels 1995-style**. ³⁴

But for a detector that must count notes from **phone photos**, I would **not** rely on synthetic augmentation alone. The public-source pool is still much stronger on **flat scans and catalog images** than on **real handheld scenes, clutter, occlusion, folds, blur, mixed-note piles, and perspective distortions**. The banknote-detection literature you surfaced also explicitly centers **smartphone-captured visible-light images**, which matches your deployment domain. ³⁶

My practical judgment is:

- **Synthetic augmentation is enough to improve class balance quickly**, especially if you build homography, scale, crop, shadow, blur, and clutter augmentations from the clean Numista templates.

- **More real data is still needed** to make the weak classes robust, especially for **KHR_20000 1995**, **KHR_20000 2017**, **KHR_50000 2001**, and **KHR_50000 1995-style**.
- The best strategy is a **hybrid** one: use open or tolerated clean scans to generate synthetic variants, then anchor each weak design family with a smaller but high-quality set of real phone photos or permission-cleared marketplace/stock images. ³⁷

1 4 5 6 7 8 9 20 https://www.nbc.gov.kh/english/about_the_bank/banknotes_in_circulation.php
https://www.nbc.gov.kh/english/about_the_bank/banknotes_in_circulation.php

2 16 31 <https://universe.roboflow.com/khmer-riel-classification-computer-vision/cambodia-currency-project>
<https://universe.roboflow.com/khmer-riel-classification-computer-vision/cambodia-currency-project>

3 12 37 <https://en.numista.com/217619>
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11 29 <https://en.numista.com/208877>
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<https://www.kaggle.com/datasets/thychantha/khmer-currency-itc-ams-gen2-c>

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